Ben Petrihos
| 703-472-9333 | blpetrihos@wm.edu | U.S. Citizen |

EDUCATION

William & Mary, Williamsburg, VA | Expected Graduation: Spring 2027
Bachelor of Arts in Government, Minor in Data Science | Major GPA: 3.94 | Cumulative GPA: 3.84
VICEROY Scholar, Department of War Cyber-Spectrum Leadership Program

EXPERIENCE

International Security Research Fellow, Futures Group – Williamsburg, VA | Academic Year 2026–2027
- Recruited by Professor Dennis A.V. Smith as one of four university research fellows; leading a team of two associates conducting a net assessment of U.S. and PRC regional competitions
- Producing international security deliverables for U.S. national security officials, focused on five-to-ten-year strategic scenarios

Research Fellow, Geopolitics of Technology Initiative – Williamsburg, VA | Academic Year 2026–2027
- Leading a team on behalf of the Global Research Institute, for the W&M / Yale policy partnership on the impact and direction of emerging technology on geopolitical competition
- Principal author of a yearlong research effort culminating in publication and presentation at national security conferences

Founding Fellow and White Cell Lead, Statecraft Simulations Group – Williamsburg, VA | 2025 – Present
- Managing the design and operations of wargames focusing on soft power, homeland security, and U.S.–China competition, including AI-modeled economic, national interest, and escalation adjudication
- Collaborated with National Defense University, J7, and J8 to develop tools and analyze simulation findings
- Presented outcomes to policymakers, senior U.S. officials, and private sector leaders
- Selected as a founding fellow under the mentorship of Major General Charlie Lyon, USAF (Ret.)

Data Science Independent Study, William & Mary – Williamsburg, VA | Fall Semester 2026
- Developing deterministic Python agents that aid in adjudication of national security simulations for academic credit, on behalf of the Chair of the Data Science Department, Professor Dan Runfola
- Serving as a research assistant to faculty on Python agent pipelines used in senior seminar adjudication
- Instructing students in how to design and build agents for national security analysis

Research Fellow, Robert M. Gates Initiative in American Statecraft – Williamsburg, VA | 2025– Present
- Developed economic, diplomatic, and information software tools to measure U.S. and partner exposure to Chinese rare earth leverage across extraction, refinement, and manufacturing
- Built an ArcGIS interactive dashboard of 32 critical minerals, including six rare earths, mapping 1,312 supply chain links across 109 countries, with cascading selectors and a 22-country map for national security simulations
- Appointed by Eric Brown, Senior Research Scientist, Global Research Institute

Department of War, Innovation for Defense – Beale AFB, CA | Langley AFB, VA | Fall Semester 2025
- Led a team tasked with developing a base-level counter drone (C-sUAS) effort: conducted 94 stakeholder interviews and site visits at Langley and Beale AFB with Security Forces, Operational Support Squadron, Explosive Ordnance Disposal, and Air Traffic Control
- Root cause analysis resulted in the stand-up of Beale AFB's first C-sUAS Common Operating Picture; identified capability gaps that drove immediate procurement of multiple sensors for layered defenses and laid the groundwork for validating a new drone-defeat framework
- Recruited and mentored by Major General Mark Matthews, USAF (Ret.)
- Identified as team leader by sponsor Captain Quinten Williams (9th OSS Beale AFB)

Research Analyst, American Legislative Exchange Council (ALEC) – Crystal City, VA | 2022

• Conducted data analysis and produced research deliverables for the Economic, Tax, and Public Affairs departments, contributing to multiple ALEC publications

• Represented ALEC at national policy conferences, briefing research to legislators and policy staff, and was credited in published research for analytic contributions

PROJECTS

- **Statecraft Simulations Group: The Petrihos Lever Index (PLI)**
  - Dashboard: Economic statecraft adjudication model for national-security simulations: a tailored AI integrated system whose deterministic engine computes quarterly macroeconomic indicator paths
  - Continuing development in Python of PLI and SDK agent pipelines for White Cell and wargaming use, including support to faculty as a research assistant and instruction of students in agent design for national security objectives
- **Geopolitics of Technology Initiative: The PRC Economic Vulnerability Index (PRCVI)**
  - Weighted composite indicator of partner-state vulnerability to Chinese economic statecraft, combining chokepoint-adjusted critical-input dependence, concentrated FDI exposure, opaque PRC lending, and services-channel dependence
  - Built a Python pipeline in pandas, NumPy, scikit-learn, and SciPy from AidData TUFF 3.0, World Bank, and UN Comtrade, with a validation harness scored against the Reynolds-Goodman CSIS inventory of PRC economic coercion cases
  - Built an interactive map with hover tooltips and click-to-select inspection, equal-radius markers, and layers for Vulnerability, CCD, HAE, OLE, SCD, plus Reynolds-Goodman coercion attempts
- **Futures Group: Buy-Side Net — Commanding Heights of Latin America**
  - Dashboard: interactive evidence map of U.S. versus PRC presence across infrastructure, scarce natural resources, and energy in Latin America and the Caribbean (GIAS Futures Group Team 2)
  - Filterable by country, subcategory, actor, and evidence class; time scrubber shows presence building year by year; Capture Pathways, Methods, and Bibliography pages document entries
- **Department of War: CTRL, ALT, Defend – Base-Level Counter Drone Readiness**
  - Root cause findings supported stand-up of Beale AFB's first C-sUAS Common Operating Picture and immediate procurement of multiple sensors for layered defenses
  - Designed a joint OSS/SF training framework with pass/fail metrics for detection and response time, and delivered a Mission Model Canvas and deployment roadmap to 9th OSS
- **American Legislative Exchange Council: Unaccountable and Unaffordable**
  - A state pension liability report surveying 290+ state-administered public pension plans across all 50 states (FY 2011–2018)

LETTERS OF RECOMMENDATION

- Maj Gen Charlie Lyon, USAF (Ret.)
- Maj Gen Mark T. Matthews, USAF (Ret.)
- Capt Quinten E. Williams, USAF
- Skip Estes, Legislative Manager, ALEC Center for Fiscal Reform
- Thomas Savidge, Research Manager, ALEC Center for Fiscal Reform